Ericsson PE Node (JAY-PE-09) Alarm Handling SOP for ALARM_5962 (Warning) - Hardware/Environmental Fault

Node: JAY-PE-09 | **Vendor:** Ericsson | **Severity:** Warning | **Alarm Code:** ALARM_5962

Description

The alarm ALARM_5962 (Warning) is triggered on the PE node JAY-PE-09 due to a hardware/environmental fault, specifically ASIC memory parity errors, line card crashes, or power supply redundancy loss. This SOP guides the troubleshooting and resolution of this failure domain on the Ericsson PE node, which is rooted in the underlying hardware architecture and protocol mechanics. The PE node's role in the network is to act as a point of presence for the service provider's customers, providing connectivity and routing services. In this scenario, the PE node's hardware or environmental conditions have failed, causing the ASIC memory parity errors, line card crashes, or power supply redundancy loss. The Ericsson PE node's design includes redundant components to ensure high availability and reliability. However, when a hardware or environmental fault occurs, the redundant components may not be able to compensate for the failed component, triggering the ALARM_5962 (Warning) alarm. This SOP outlines the diagnostic and resolution steps to isolate and fix the root cause of the fault, restoring the PE node to a stable and operational state.

Pre-requisite Checks

- Verify that the Jayanagar Metro Site is accessible and that remote hands are dispatched to the site.

- Check the redundant link status to ensure that it is operational and not contributing to the fault.

- Review the alarm history to determine the timestamp and duration of the ALARM_5962 (Warning) alarm.

Potential Root Causes

Cause	Probability
ASIC memory parity error	High
Line card crash	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

```
Command: admin:show alarm detail ALARM_5962, Expected Output: Alarm details for ALARM_5962 (Warning), Description: This command displays detailed information about the ALARM_5962 (Warning) alarm, including the timestamp, duration, and affected components.
```

```
Command: admin:show system status, Expected Output: System status information, Description: This command displays the system status, including the operational status of the redundant components and any fault indications.

Command: admin:show hardware status, Expected Output: Hardware status information, Description: This command displays the hardware status, including the operational status of the ASIC, line cards, and power supplies.
```

Resolution Steps

Step 1: Step: Replace the faulty ASIC, Command: admin:replace ASIC ASIC_ID, Description: Replace the faulty ASIC with a new one to restore the PE node's functionality.

Step 2: Step: Replace the faulty line card, Command: admin:replace line card LINE_CARD_ID, Description: Replace the faulty line card with a new one to restore the PE node's functionality.

Step 3: Step: Replace the faulty power supply, Command: admin:replace power supply POWER_SUPPLY_ID, Description: Replace the faulty power supply with a new one to restore the PE node's functionality.

Post-Check Verification

```
Command: admin:show alarm detail ALARM_5962, Expected Output: Alarm cleared, Description: Verify that the ALARM_5962 (Warning) alarm has been cleared.

Command: admin:show system status, Expected Output: System status: normal, Description: Verify that the system status is normal and there are no fault indications.

Command: admin:show hardware status, Expected Output: Hardware status: normal, Description: Verify that the hardware status is normal and there are no fault indications.
```

Roll-back Procedures

- Step: Revert the replacement operation, Command: admin:revert replace ASIC ASIC_ID, Description: Revert the replacement operation to restore the original faulty component.

- Step: Revert the replacement operation, Command: admin:revert replace line card LINE_CARD_ID, Description: Revert the replacement operation to restore the original faulty line card.

- Step: Revert the replacement operation, Command: admin:revert replace power supply POWER_SUPPLY_ID, Description: Revert the replacement operation to restore the original faulty power supply.